

# Lab: Exploring Communication in Microstructural Evolution

## Objectives

**Learning Goal:** Apply the Active Inference concept of communication to a concrete metallurgical scenario using image analysis lab techniques.

This is a **image analysis lab exercise** designed for materials scientists and engineers.

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## Part 1: System Definition (10 min)

Choose a metallurgical system to analyze. Suggested options:

- Fe-0.4wt%C steel undergoing austenitization and cooling
- Al-4wt%Cu alloy during precipitation aging
- Ti-6Al-4V produced by laser powder bed fusion
- Pure copper undergoing recrystallization after cold work

*Write your response here*

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## Part 2: Communication Analysis (25 min)

### 2A: Identify the Active Inference Components

Map your chosen system onto the Active Inference framework in the context of communication:

| Component                        | Your System |
|----------------------------------|-------------|
| System boundary (Markov blanket) |             |
| Internal states                  |             |
| External states                  |             |
| Sensory states                   |             |
| Active states                    |             |

*Write your response here*

## 2B: Communication Dynamics

Describe how communication operates in your system:

- What is the driving force (free energy gradient)?
- What is the timescale?
- What characterization technique would you use to observe it?

*Write your response here*

## Part 3: Quantitative Exercise (20 min)

Perform a simple calculation or simulation related to communication in your system:

- If computational: use Python with PyCalphad, NumPy, or similar

- If analytical: use thermodynamic relations (Gibbs energy, diffusion equation, nucleation barrier)

*Write your response here*

### Part 4: Reflection (10 min)

1. How did the Active Inference framework change your understanding of communication in this system?
2. What prediction error exists between the equilibrium model and the real behavior of your system?
3. How could you reduce this prediction error through better characterization or modeling?

*Write your response here*

### Summary

| Analysis Component                 | Key Finding |
|------------------------------------|-------------|
| System analyzed                    |             |
| Communication mechanism identified |             |
| Driving force                      |             |
| Main prediction error              |             |
| Recommended next characterization  |             |

*Write your response here*